

MyGPT Design and Architecture

ASE 285 Individual Project

Project Overview

MyGPT is a React and Electron chatbot application.

It uses the OpenAI API for AI responses.

It stores user data in localStorage.

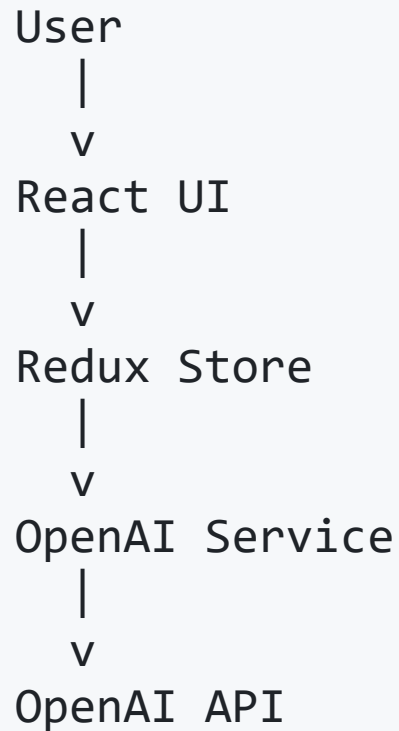
Main Goal

The goal is to improve a ChatGPT clone prototype.

The app adds user-friendly features:

- Help
- UI settings
- Search
- Bookmarks
- Documents
- Conversation title editing

High-Level Architecture



Main Folders

```
src/  
  src/  
    Dashboard/  
      Chat/  
      Help/  
      Sidebar/  
      dashboardSlice.js  
      Dashboard.jsx  
  services/  
    openaiService.js
```

React UI Layer

The React UI handles the screens and user actions.

Important parts:

- `Dashboard.jsx` : main app screen and feature pages
- `Sidebar.jsx` : conversation list and sidebar buttons
- `Chat.jsx` : chat area
- `NewMessageInput.jsx` : message input and document picker
- `HelpOverlay.jsx` : help guide overlay

State Management

Main state:

- conversations
- selected conversation ID
- loading state
- error state

OpenAI API Flow

1. The user writes a prompt.
2. The app creates a user message.
3. The app adds selected document text if needed.
4. `sendConversationMessage` sends the prompt to `openaiService`.
5. The OpenAI response is added to the conversation.
6. The conversation is saved in `localStorage`.

Document Flow

Documents are saved in localStorage.

The user can add, edit, delete, and select documents.

When documents are selected, their text is added to the prompt as reference context.

Bookmark Flow

Each message can have a `bookmarked` value.

When the user clicks the bookmark button, the app toggles this value.

The Bookmarks page collects bookmarked messages from all conversations.

Search Flow

The search page checks saved conversations.

It compares the keyword with message text.

Matching conversations are shown as results.

The user can open a conversation from the result list.

Data Model

Conversation

- id
- title
- messages[]

Message

- id
- content
- aiMessage
- bookmarked

Document

- id
- content
- savedAt

Local Storage Keys

- `conversations`
- `documents`
- `uiFontSize`

Design

The app keeps the design simple.

Most features are in the dashboard area because they are user-facing chat features.

Future Improvements

- Add cloud database support.
- Improve mobile layout.
- Add more customizing features.